K S Suraksha Jain | CL24M015 | Indian Institute of Technology Madras | GitHub | LinkedIn |

EDUCATION AND SCHOLASTIC ACHIEVEMENT

Education	Institute	% / CGPA	Year
M Tech, Clinical Engineering	Indian Institute of Technology Madras, Tamil Nadu	7.63 CGPA	2026
B E, Biotechnology	Nitte Mahalinga Adyantaya Memorial Institute of Technology	8.39 CGPA	2022
Class XII (State)	Karkala Jnanasudha PU College, Udupi, Karnataka	93.66%	2018
Class X (State)	Swasthi Shree Nemisagara Varneeji, High school, Udupi, Karnataka	96.16%	2016

- Secured **AIR 696** (All India Rank) in GATE 2024 – Biotechnology (BT) out of 17078 students.

KEY PROJECTS

Course-CMC Vellore* | **COHMET - Cognitive and Hand Motor skill Enhancement Trainer** | *(Mar-Jun 25)*

- Designed & developed a smart electronic pegboard for **cognitive and motor rehabilitation** of **stroke and spinal cord injury patients**.
- Designed **mechanical structures** for sensor integration & placement of addressable LEDs using **SketchUp** and fabricated using **3d printer**.
- Built **custom app** to control the device via HC05 module and store the patient data to **google sheets API** accessible via **custom website**.
- Successfully tested the device on **5 different patients** at CMC* Vellore rehabilitation centre with **validation from doctors and therapists**.

Course-AM5011-IITM | **Virtual Rebirth- Visualization of Lost Monuments Using VR* and custom navigating hardware.** | *(Oct-Dec 24))*

- Designed a Unity Based VR game for **visualizing old lost structure** of Konark Sun temple modelled using Sketchup with literature inputs.
- Developed a detailed 3D model in SketchUp from Google Maps and archaeological references, and seamlessly **rendered it in Unity.**
- Learnt and applied the concepts of vection, XR Rig, **rendering pipeline**, **collision detection physics**, **spatial audio** to realise the project.

*Course-SCTIMST** | **Modification of patient lift for locomotor parkinsonism rehabilitation** | *(Jul-Aug 2025)*

- Researched muscle rigidity, trunk bending restrictions, and actuation principles to develop an optimized **rehabilitation chair** design.
- **Designed the complete rehabilitation chair model** on the top of already available rehabilitation lift in **Creo**, determining actuator lengths and **proper joints and gears** to achieve the targeted motion angles to improve spine posture and reduce rigidity in back muscles.
- Performed **mesh sizing analysis**, **optimized mesh parameters**, and structural stress analysis using **Ansys** to **optimize the design**.

Personal Project | **Diabetic readmission prediction for high-sensitivity cleaning** | *(Jul-Aug 2025)*

- **Optimized LightGBM** for imbalanced diabetes readmission (89:11) using class weighting and threshold for minority class prediction.
- **Tuned key hyperparameters-learning rate, depth, leaves and regularization** and evaluated using **precision, recall, F1-score**.
- Achieved ~97% recall by selecting a custom decision threshold, balancing the recall–precision trade-off for **high-sensitivity screening**.

Personal Project | **Python based Banking Management System** | *(Jan-Feb 2025)*

- Built a Python-based Banking Management System using **OOP** and **Tkinter GUI** for account management, transactions, and reporting.
- Designed a **modular structure** with separate files for core functions, all linked to a single **SQLite3 database** for persistent storage.
- Added **admin authentication**, **input validation**, and **exception handling** for secure, reliable, and real-time banking operations.

Personal Project | **ECG Classification using Autoencoder** | *(Mar-Apr 25)*

- **Pre-processed** MIT-BIH ECG beats and **normalized** them to a uniform scale, ensuring consistent input for model training.
- Built a **1D convolutional autoencoder** to compress ECG beats into **64 features** with **low reconstruction error (MSE ≈ 4e-5**).
- Classified Normal vs. Abnormal ECG beats using latent features with Random Forest, achieving **97.8%** accuracy and **0.993 ROC-AUC**.

Course-AM5510-IITM | **Modelling and its evaluation for Time Series Data using (AR, MA, ARMA, ARIMA) models.** | *(Sept-Nov 24)*

- Modelled and evaluated **time series data** like ECG and EMG using different models like Auto Regression, Moving Average, ARMA, ARIMA
- Analysed the data using **time domain**, **frequency domain** methods. Used **NumPy, Pandas & Matplotlib** to implement those methods.

*Course-SCTIMST** | **Regulation strategy development for heart valve** | *(Aug 25- present)*

- Developing regulatory strategies for heart valves by aligning documentation with **international standards** to support successful approvals.
- Conducting **risk assessments** and **gap analyses** to identify compliance issues, mitigate challenges, and ensure adherence to standards.
- Collaborating with cross-functional teams in regulatory, R&D, and quality to prepare submissions and streamline approval processes.

Personal Project | **Automated Pneumonia Classification from Chest Radiographs Using DenseNet201** | *(May-Aug 25)*

- Developed a **DenseNet201**-based **TensorFlow/Keras** model for pneumonia detection using Kaggle chest-Xray pneumonia dataset.
- Applied preprocessing with **RGB conversion, normalization, and augmentation** to X-ray images, optimized with GPU in **Google Colab.**
- Achieved ~90% training and ~89% test accuracy for accurately classifying normal vs. pneumonia cases.

Relevant Courses, Tools, and Skills

- Linear Algebra
- Biostatistics
- Computer Aided Design
- Basic Programming
- Numerical Methods
- Heat and Mass Transfer
- Bioprocess principles and calculations
- Clinical Attachment

- Virtual Reality Engineering
- Biomedical Signal and Systems
- Biomedical Sensors and Measurement
- Biomechanics
- Biomedical Imaging

- Quality system management
- Biomaterials
- Design Tools for Clinical Engineering
- Medical Device Technology
- Critical Care Instrumentation
- Physiology and Anatomy

- **Language**-Python
- **Software**- Creo, ANSY, VS code, Figma, Sketchup, Arduino IDE, Git, GitHub, Unity, Jupyter notebook, Google Colab
- **Skills**- Web Dev (basic), GUI, OOPS, 3D Modelling, Graphics Design, Prototyping, 3D printing & product development, documentation,
- **Framework**- Tkinter, NumPy, Pandas, Matplotlib, OpenCv, SQLite, Flask, TensorFlow, Keras. Google Cloud API, MediaPipe

Conferences Attended

Conference
- Attended the Winter Symposium on "Health Data & AI" at CMC Vellore.
- Internation conference on Nurturing innovative technological trends in engineering bioscience. NMAMIT

Hobbies and Others

Hobbies-Bharatnatyam, Hiking, Cooking, Yoga, Swimming, reading books. **Languages**- English, Hindi, Tulu, Kannada

* Christian Medical College, Virtual Reality, Sri Chitra Thirunal Institute of Medical Science and Technology